

EX 2: Bias Audit of Pre-Trained Word Embeddings Using WEAT

Python Program

```
!pip install gensim numpy

import numpy as np
from gensim.models import KeyedVectors

embedding_path = "/content/drive/MyDrive/GoogleNews-vectors-negative300.bin"

model = KeyedVectors.load_word2vec_format(embedding_path, binary=True)

X = ["man", "male", "boy"]
Y = ["woman", "female", "girl"]

A = ["career", "business", "office", "salary"]
B = ["family", "home", "children", "marriage"]

def cosine_similarity(w1, w2):
    return np.dot(w1, w2) / (np.linalg.norm(w1) * np.linalg.norm(w2))

def association(w, A, B):
    assoc_A = [cosine_similarity(model[w], model[a]) for a in A]
    assoc_B = [cosine_similarity(model[w], model[b]) for b in B]
    return np.mean(assoc_A) - np.mean(assoc_B)

def weat_score(X, Y, A, B):
    score_X = [association(x, A, B) for x in X]
    score_Y = [association(y, A, B) for y in Y]
    return sum(score_X) - sum(score_Y)

def effect_size(X, Y, A, B):
    assoc_X = [association(x, A, B) for x in X]
    assoc_Y = [association(y, A, B) for y in Y]
    mean_diff = np.mean(assoc_X) - np.mean(assoc_Y)
    std_dev = np.std(assoc_X + assoc_Y)
    return mean_diff / std_dev

score = weat_score(X, Y, A, B)
effect = effect_size(X, Y, A, B)

print("WEAT Score:", score)
print("Effect Size:", effect)
```

Sample Output

```
WEAT Score: 1.87
Effect Size: 0.62
```